[ITA0328 - Mobile computing

Concept mapping

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Handoff mechanisms in cellular networks

Goal: maintain uninterrupted communication during mobility

1. HARD HANDOFF (Break-before-make)

- old link released before new link is established.
- connects to one cell at a time.
- common in 2G (GSM) 4G (LTE)

Suitability

Higher drop risk; suitable for low mobility.

2. Soft handoff (make-Before-Break)

- New link established before old link is released.
- connects to multiple cells simultaneously.
- common in 3G (wcdma, CDMA)

suitability

★★★★☆

Very low drop probability; ideal for high mobility.

3. Handoff decision parameters.

- RSS / signal strength.
- SIR / SINR (interference)
- velocity of MS.
- Hysteresis margin.
- time to trigger (TTT)
- cell load QoS, etc.

Role

Determine when and to which cell handoff should occur for optimal performance.

comparison	Hard hand off	Soft Handoff	Decision
connections at a time	1	multiple	N/A
call drop probability	Higher	very Low	minimized if well tuned
signaling overhead	Low	High	Depends on algorithm
Best suitable	Low mobility simple networks	High mobility seamless QoS	optimizes hand off

Conclusion:- Soft handoff is best for uninterrupted communication; hard handoff is simpler but risky. Handoff decision parameters ensure the right handoff at the right time for better QoS.